Jinghao Wen

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RESEARCH INTERESTS

I aspire to build efficient, reliable, and trustworthy computing systems for next-generation machine learning. My research interests lie in computer architecture, machine learning systems, and system reliability, with a focus on improving AI robustness under hardware faults, soft errors, and resource constraints while balancing performance and efficiency.

EDUCATION

Villanova University, Villanova, PA 09/2024-present
Ph.D in Computer Engineering

Central China Normal University, Wuhan, China 09/2020-06/2024
Bachelor of Engineering in Computer Science and Technology, Core GPA: 87.1/100, Ranking: 3/182

- Honors: National Scholarship for Undergraduates (2024); Golden Scholarship for Merit Students (2021 & 2022)
- Key coursework: Operating System, Data Structures, Database, Algorithms Design and Analysis, Machine Learning

PUBLICATION

"Improving Efficiency and Reliability of Computer Vision Models in Real-World Deployment with MX Quantization"; **Jinghao Wen**, Ruixuan Wang, Shaohuang Wang, Xun Jiao; *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026*

"Learning Fingerprints for Medical Time Series with Redundancy-Constrained Information Maximization"; Huayu Li, ZhengXiao He, Xiwen Chen, Jingjing Wang, siyuan tian, **Jinghao Wen**, Ao Li; *International Conference on Machine Learning (**ICML**), 2026*

"Martingale Foresight Sampling: A Principled Approach to Inference-Time LLM Decoding"; Huayu Li, ZhengXiao He, Siyuan Tian, **Jinghao Wen**, Ao Li; *The European Chapter of the ACL (**EACL**), 2026*

"MILEAM: Modeling Input-Aware Logic Errors of Approximate Multipliers using Machine Learning"; **Jinghao Wen**, Dongning Ma, Xun Jiao; *IEEE International Symposium on Circuits and Systems (**ISCAS**), 2026*

"Beyond Sampling: Classwise Loss Optimization for Imbalanced Deep Learning Recommendation Systems"; Xulei Zhang*, **Jinghao Wen***, Xun Jiao; *IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**), 2026*

"HDFlow: A Multi-layer Design and Evaluation Framework for Hyperdimensional Computing"; **Jinghao Wen**, Dongning Ma, Sizhe Zhang, Xun Jiao; *The International Joint Conference on Neural Networks (**IJCNN**), 2026*

"NeuroHD-RA: Neural-Distilled Hyperdimensional Model With Rhythm Alignment"; Zhengxiao He*, **Jinghao Wen***, Huayu Li, Siyuan Tian, Xin Yang, Ao Li; *The Annual International Conference of the IEEE Engineering in Medicine and Biology Society (**EMBC**), 2026*

" Proof-of-Useful-Work Blockchain for Trustworthy Biomedical Hyperdimensional Computing"; **Jinghao Wen**, Dongning Ma, Sizhe Zhang, Hasshi Sudler, Xun Jiao; *IEEE Biomedical Circuits and Systems Conference (**BioCAS**), 2025*

"A Comparative Study of Structured and Narrative EHR Data for 30-Day Readmission Risk Assessment"; Zhengxiao He, Huayu Li, Siyuan Tian, **Jinghao Wen**, Geng Yuan, Ao Li; ***Electronics**, 2025*

"WiFi-based non-contact human presence detection technology"; Yang Zhang, Xuechun Wang, **Jinghao Wen**, Xianxun Zhu; ***Scientific Reports**, 2024*

RESEARCH EXPERIENCE

Research Assistant, University of Florida, SmartSystems Lab 06/2023-08/2023
Advised By: **Prof. Christophe Bobda** *Gainesville, FL*

- For multi-camera system, realized the transformation and calibration among world coordinate, camera coordinate, image coordinate, pixel coordinate. Achieved camera pose estimation and triangulation.
- Applied the Canny and Sobel in edge detection, compared and applied Surf and Sift in feature detection and keypoints matching. Added adaptive filters for a more smooth effect on noise and a less smooth effect on edge.

- Improved the anti-noise performance of the Sobel operator by using the weighted nuclear norm minimization (WNNM) image denoising algorithm.

Team Leader, Kaggle Machine learning Competition 11/2022-02/2023
*Silver Medal at Kaggle Competition, **top4%**, 2587 teams worldwide*
- *Competition topic*: building a multi-objective session-based recommender system to predict users' next clicks, cart additions, and orders in an e-commerce platform.
- Used multiple recall methods to identify the top 200 items that are most likely to be clicked/favored/purchased.
- Established a word2vec model, vectorized and coded the products. Established a product similarity matrix, based on the user's click behavior sequence from the real data in the German OTTO e-commerce platform.
- Constructed a series of characteristics of items, users, and item-user dimensions based on the recalled items.
- According to the status of the product, labeled the recalled data with 0 or 1. Established a LGB binary classification model, and selected 20 product IDs with the highest prediction probability.

Undergraduate Researcher, Central China Normal University, Smart Robot Lab 12/2020-03/2023
- *Research Topic*: developing an intelligent humanoid robot system with ROS-based simulation, embedded deployment, computer vision, gait planning, and servo-motor motion control.
- Simulated robot in ROS and deployed algorithms on Jetson Nano. Built a URDF model of the robot, developed robot's gait planning, path planning and conduct simulation in Rviz and Gazebo.
- Implemented OpenPose-based human posture recognition with 91.7% accuracy and converted joint skeleton distance and angle information into servo motor control signals, achieving an 88.6% action synchronization rate.

Undergraduate Researcher, Central China Normal University, Smart Robot Lab 05/2021-06/2022
- *Research Topic*: designing an Raspberry Pi-based intelligent quadruped robot with ROS-based simulation, SLAM map planning, UWB localization, wireless communication, multi-sensor fusion, and embedded object recognition.
- Built a Zigbee communication module to realize communication between robots. Enabled the robot's movement and obstacle avoidance by integrating the data of voice recognition, infrared laser, gyroscope and other sensors.
- Trained the Yolo V5 model in virtue of the products' image data autonomously collected by the camera and deployed the program to the Raspberry Pi to implement the robot's real object recognition.

SELECTED PROFESSIONAL EXPERIENCE

Co-Funder & Machine Learning Engineer 10/2025-present
Lagram Technologies *Shanghai & New York, NY*
- Co-founded an AI-driven compliance automation startup for Testing, Inspection, and Certification (TIC) workflows, developing systems for regulatory reasoning, report generation, and customer management.
- Built machine-learning-based compliance reasoning systems combining hybrid document retrieval, regulatory knowledge graphs, and structured report generation with traceable evidence. Designed standard-agnostic architectures that separates regulatory clauses from workflow logic, enabling rapid adaptation across different national standards and product categories.
- Developed an automated annual-fee management system to summarize customer service records, calculate annual fees, generate client quotations, and streamline email-based customer communication.
- Supported early commercial deployment with international TIC companies (Intertek, Dekra) validating the system through real industrial compliance-reporting and customer-management workflows.

Machine Learning Engineering Intern (Research-Oriented) 08/2025-12/2025
Internet Think Tank, Inc *Woodland Hills, CA*
- Developed HDCoin, a proof-of-useful-work blockchain framework that transforms conventional mining into useful training of trustworthy biomedical Hyperdimensional Computing (HDC) models.
- Designed HDC-based mining and verification pipelines where distributed miners train, validate, reproduce, and verify biomedical learning models using nonce-generated item memories and HDC hyperparameter configurations.
- Conducted design-space exploration across HDC dimensionality, retraining iterations, and retraining rates on four biomedical datasets, including UCIHAR, BreastMNIST, PneumoniaMNIST, and NoduleMNIST3D.
- Evaluated adaptive mining difficulty and fairness mechanisms, showing how HDC configurations affect computational effort, model performance, and miners' winning probability.

Project Product Manager 03/2024-05/2024

Momenta (Autonomous Driving Company) *Shanghai, China*

- Supported end-to-end project management for autonomous driving R&D projects, tracking milestones, timelines, and cross-functional resources across algorithm, software, and testing teams.
- Participated in real-world autonomous driving road tests on elevated highways in Shanghai, accumulating **3,000+ miles** of on-road testing under real traffic conditions.
- Assisted in product requirement definition, issue analysis from testing, and technical documentation to facilitate efficient development and delivery.
- Collaborated with product and engineering teams to translate business and customer needs into actionable technical and product requirements. Acted as a liaison between algorithm teams and external clients, aligning technical details, clarifying requirements, and supporting solution delivery.

Software Development Engineer Intern 10/2023-12/2023

Tencent *Shenzhen, China*

- Built responsive frontend features and reusable UI components for internal web platforms using Vue.js, TypeScript, and JavaScript. Developed reusable page components, form modules, data tables, and dashboard views with Vue Router and Axios, improving feature development efficiency.
- Integrated RESTful APIs with backend services, handled request states, permission-based rendering, form validation, and frontend error handling in production workflows.
- Participated in the full development process, including requirement review, UI implementation, code review, Git-based collaboration, testing, debugging, and deployment support.

HONORS & AWARDS

Google Cloude Research Credits Scholarship 2026

Yongping'89 and Fei Lin'89 Gu Endowed Graduate Engineering Fellowship 2025

China National Scholarship for Undergraduates 2024

Silver Medal at OTTO - Multi-Objective Recommender System Competition (**Kaggle top4%**, 2587 teams) 2023

Second Prize at the national level in the 16th **China College Students Computer Design Competition** 2023

First Prize at the national level in 2022 **China Robot Competition & RoboCup China Open** (dancing robot) 2022

Second Prize at the national level in 2022 **China Robot Competition & RoboCup China Open** (creative robot) 2022

TEACHING & MENTORSHIP

Villanova University: ECE1261,Engineering Programming & Applications Lab, Teaching Assistant 2026

Villanova University: Intern Mentor: Xianglong Yu (UPenn, graduate student) 2026

CCNU: Advanced Language Programming Experimental Course, Teaching Assistant 2024

CCNU: Data Structure, Teaching Assistant 2023

TECHNICAL SKILLS & LANGUAGES

Programming Languages: Python, C/C++, JavaScript, Java, x86 Assembly, HTML

AI & Big Data: SQL, MySQL, Pytorch, Linux, Tensorflow, NumPy, Pandas, Jupyter

Embedded hardware: Jetson Nano, Raspberry Pi, Arduino, ROS

Tools: Git, SolidWorks, 3ds Max, Photoshop, Premiere Pro

Languages: Chinese (native), English (IELTS:7.0)

ACTIVITIES AND OTHER EXPERIENCES

Reviewer: AAAI, TCAD, ICASSP, IJCNN, MWSCAS, EMBC

Volunteer: China Robot Competition and RoboCup China Open